Go

Small building blocks for concurrent services

THE MENTAL MODEL

Many independent workers can communicate through channels.

REQUEST → GOROUTINES → CHANNELS → SERVICE

READ THE CODE

```
package main
import "fmt"
func main() {
    values := []int{10, 20, 30}
    for _, x := range values {
        fmt.Println(x)
    }
}
```

FOUR CLUES TO READ IT

- package groups related Go files.
- := declares and initializes a local variable.
- range visits a collection; _ ignores the index.
- fmt.Println prints each value.

WHAT MAKES IT DIFFERENT

- Goroutines are lightweight concurrent functions; channels can coordinate them.
- Simple tooling and static binaries are convenient in service deployment.

WHERE IT FITS

Network servers, command-line tools, cloud infrastructure.

WATCH OUT

Concurrency needs coordination; goroutines do not remove data races.

TRY ALONE • 60 SEC

Which of the two values from range is ignored, and why?